

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belgaum)***Eighth Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

May - June 2023

19BTC210 - PHARMACEUTICAL BIOTECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Give an account on excipients used in parenteral formulations with reference to preservatives and osmotic agents.	10	L*2	1	2,1
	b) Explain freeze drying of proteins.	10	L2	2	3,1
2.	a) Elaborate protein delivery by microencapsulated secretory cells.	10	L2	2	2,1
	b) Describe colloidal particulate system for targeted delivery of proteins.	10	L2	2	3,1
3.	a) Write an essay on site specific delivery of protein drugs.	10	L2	2	2,1
	b) Add a note on i) Viral decontamination ii) Closed loop system.	5+5	L2	1	2,1
Unit – II					
4.	a) Explain soluble preparation and long acting insulin formulations.	10	L2	3	1,4,1
	b) Discuss on hGH structure and its pharmacology.	10	L2	3	1,4,2
5.	a) Give a detailed account on vaccines.	10	L2	4	2,1,3
	b) Write a note on i) GM-CSF ii) t-PA.	5+5	L2	4	2,1,3
6.	a) Explain monoclonal antibody and its therapeutic applications.	10	L2	4	2,1
	b) Comment on i) recombinant factor VIII and its pharmacology, ii) Tenecteplase.	5+5	L2	4	3,1
Unit – III					
7.	a) Explain FDA guidelines on compatibility.	10	L2	5	1,2
	b) Give an account on challenges of regulatory issues.	10	L2	5	2,1
8.	a) Discuss on clinical studies.	10	L2	5	1,2
	b) Describe immunogenicity with reference to EMEA document.	10	L2	5	1,4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
